

price momentum and better transaction prices. The trading cost standard error term is comprised of price volatility, liquidity risk, and parameter estimation error from the market impact model.

Cost versus Profit and Loss

There is not much consistency in the industry regarding the terminology or sign to use when measuring and forecasting costs. Many participants state cost as a positive value while others state cost as a negative value. For example, some participants refer to a positive cost of +30 bp as underperformance and a negative cost of −30 bp as outperformance (savings). Others treat this metric in the opposite way with the +30 bp indicating better transaction prices and −30 bp indicating worse transaction prices.

To avoid potential confusion, our “Cost” and “Profit and Loss” terminology throughout the text will be as follows:

A “**Cost**” metric will always use a positive value to indicate underperformance and a negative value to indicate better performance. For example, a cost of 30 bp indicates less favorable execution than the benchmark and −30 bp cost indicates better performance than the benchmark.

A “**Profit and Loss**” or “**PnL**” metric will always use a negative value to indicate underperformance and a positive value to indicate better performance. For example, a PnL of −5 bp indicates less favorable execution than the benchmark and a PnL of +5 bp indicates better performance compared to the benchmark.

IMPLEMENTATION SHORTFALL

Implementation shortfall (IS) is a measure that represents the total cost of executing the investment idea. It was introduced by Perold (1988) and is calculated as the difference between the paper return of a portfolio where all shares are assumed to have transacted at the manager’s decision price and the actual return of the portfolio using actual transaction prices and shares executed. It is often described as the missed profiting opportunity as well as the friction associated with executing the trade. Many industry participants refer to implementation shortfall as slippage or simply portfolio cost.

Mathematically, implementation shortfall is written as:

$$IS = \text{Paper Return} - \text{Actual Return} \quad (3.1)$$